

A Two-Sided Strong-Exposure Obstruction to Numerical Index One

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Source Problem

The source paper is Pal–Roy–Sain, *A characterization of Banach spaces with numerical index one*, arXiv:2604.03690 [1]. On page 7, after recalling necessary conditions of Lopez–Martin–Paya [2], the authors write that it is open whether the real reflexive finite-dimensionality theorem remains true for complex Banach spaces:

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- i) $|\chi(x^*)| = 1$ for every $\chi \in E_{B_{X^{**}}}$ and every weak*-denting point $x^* \in B_{X^*}$.
- ii) $|x^*(x)| = 1$ for every $x^* \in E_{B_{X^*}}$ and every denting point $x \in B_X$.

Let us also mention here that in the same paper [14], it has been shown that a reflexive real Banach space with numerical index one must be finite-dimensional. It is an open problem whether the same result also holds true for complex Banach spaces. In addition to this, it has been proved in [5] that M-spaces, L-spaces, and their isometric preduals have numerical index one. We refer to [11] for more information on this topic. Before presenting the promised characterization of Banach spaces with numerical index one, we recall the definition of a spear element [1, 8] in a Banach space X . An element $z \in X$ is called a *spear element* if for every $x \in X$ there is a modulus one scalar t for which $\|z + tx\| = 1 + \|x\|$ holds.

Theorem 2.10. *Let X be a Banach space. Then the following statements are equivalent*

- (a) $n(X) = 1$.

- (b) $B_{X^{**}} = \overline{\text{weak}^*} \{x^* \odot y : x^* \in E_{B_{X^*}}, y \in S_X, |x^*(y)| = 1\}$

In other words, the open problem is whether every complex reflexive Banach space with numerical index one must be finite-dimensional.

Proof Intuition

The numerical index condition $n(X) = 1$ says that every operator has numerical radius equal to its operator norm. A two-sided strongly exposed support pair gives enough local rigidity to defeat this for a rank-one operator. If x^* strongly exposes x , then any numerical-radius test vector that nearly norms the rank-one functional factor x^* must be close to the circle orbit of x . If, at the same time, x strongly exposes x^* in the dual ball, then the corresponding support functionals must be close to the circle orbit of x^* . Choosing the rank-one operator $Tz = x^*(z)y$ with $y \in \ker x^*$ forces those same support functionals to be almost zero on y , contradicting numerical radius one.

Thus a complex reflexive counterexample, if it exists, cannot have even one such two-sided smooth/rotund support point. The remaining difficulty in the full open problem is that reflexivity supplies denting phenomena, but not automatically this matched two-sided strong exposure for the original norm.

Definitions

Let X be a complex Banach space. The numerical radius of $T \in L(X)$ is

$$w(T) = \sup\{|x^*(Tx)| : x \in S_X, x^* \in S_{X^*}, x^*(x) = 1\},$$

and the numerical index is

$$n(X) = \inf\{w(T) : T \in L(X), \|T\| = 1\}.$$

Definition 1. Let $x \in S_X$ and $x^* \in S_{X^*}$ satisfy $x^*(x) = 1$. We say that (x, x^*) is a two-sided strongly exposed support pair if:

1. whenever $z_\alpha \in B_X$ and $\operatorname{Re} x^*(z_\alpha) \rightarrow 1$, then $\|z_\alpha - x\| \rightarrow 0$;
2. whenever $f_\alpha \in B_{X^*}$ and $\operatorname{Re} f_\alpha(x) \rightarrow 1$, then $\|f_\alpha - x^*\| \rightarrow 0$.

The first condition says that x^* strongly exposes B_X at x . The second says that x , viewed as an element of X^{**} , strongly exposes B_{X^*} at x^* .

Main Partial Result

Theorem 1. Let X be a complex Banach space with $\dim X \geq 2$. If X has a two-sided strongly exposed support pair, then $n(X) < 1$.

Proof. Let (x, x^*) be such a pair. Since $\dim X \geq 2$, choose $y \in S_X \cap \ker x^*$. Define the rank-one operator

$$Tz = x^*(z)y \quad (z \in X).$$

Then $\|T\| = 1$. We claim that $w(T) < 1$.

Suppose, toward a contradiction, that $w(T) = 1$. By the definition of the supremum, for each n there are $u_n \in S_X$ and $u_n^* \in S_{X^*}$ such that $u_n^*(u_n) = 1$ and

$$|u_n^*(Tu_n)| = |x^*(u_n)| |u_n^*(y)| \rightarrow 1.$$

Hence both $|x^*(u_n)| \rightarrow 1$ and $|u_n^*(y)| \rightarrow 1$.

Put $\lambda_n = \overline{x^*(u_n)}/|x^*(u_n)|$ after discarding finitely many terms for which $x^*(u_n) = 0$, and set

$$v_n = \lambda_n u_n, \quad v_n^* = \overline{\lambda_n} u_n^*.$$

Then $v_n^*(v_n) = 1$, $|v_n^*(y)| = |u_n^*(y)| \rightarrow 1$, and

$$x^*(v_n) = |x^*(u_n)| \rightarrow 1.$$

Since x^* strongly exposes B_X at x , we have $\|v_n - x\| \rightarrow 0$. Consequently,

$$|v_n^*(x) - 1| = |v_n^*(x) - v_n^*(v_n)| \leq \|x - v_n\| \rightarrow 0.$$

Thus $\operatorname{Re} v_n^*(x) \rightarrow 1$. Since x strongly exposes B_{X^*} at x^* , it follows that $\|v_n^* - x^*\| \rightarrow 0$. But $x^*(y) = 0$, so

$$|v_n^*(y)| \leq \|v_n^* - x^*\| \|y\| + |x^*(y)| \rightarrow 0,$$

contradicting $|v_n^*(y)| \rightarrow 1$. Therefore $w(T) < 1$, and since $\|T\| = 1$, we get $n(X) \leq w(T) < 1$. $\square$

Corollary 1. *Let X be a complex Banach space with $n(X) = 1$ and $\dim X \geq 2$. Then X has no two-sided strongly exposed support pair.*

Corollary 2. *If a complex Banach space X of dimension at least two has a point $x \in S_X$ which is locally uniformly rotund and Frechet smooth, and if its unique support functional x^* satisfies the corresponding dual strong-exposure condition above, then $n(X) < 1$. In particular, a uniformly convex and uniformly smooth complex Banach space of dimension at least two has numerical index strictly smaller than one.*

Proof. Local uniform rotundity at x implies that the unique support functional x^* strongly exposes B_X at x . Frechet smoothness at x is the standard duality condition giving strong exposure of B_{X^*} at x^* . Apply the theorem. The uniformly convex and uniformly smooth case satisfies these hypotheses at every unit vector. $\square$

Scope and Limitations

This is a partial obstruction, not a solution to the full open problem. It proves that any infinite-dimensional complex reflexive Banach space with numerical index one would have to avoid matched two-sided strong exposure in its original norm. Reflexive spaces have the Radon–Nikodym property and denting structure, but denting alone is weaker than the two-sided strong exposure used in the rank-one argument above.

Novelty and Search Notes

The lightweight run indexes had no entry for arXiv:2604.03690 or this two-sided strong-exposure obstruction. Web searches for the exact complex-reflexive numerical-index-one formulation mainly returned the source paper itself. Nearby run memory contains numerical-radius and Daugavet/lushness packets, but none addresses this source problem or this rank-one obstruction.

Verification Notes

The proof is a direct contradiction argument with the rank-one operator $Tz = x^*(z)y$. The only external facts used are the source formulation of the open problem and standard duality between local uniform rotundity/Frechet smoothness and strong exposure. No numerical computation is involved. The human reviewer should mainly check the phase normalization in the complex numerical-radius argument and the exact hypotheses in the corollaries.

References

- [1] S. Pal, S. Roy, and D. Sain, *A characterization of Banach spaces with numerical index one*, arXiv:2604.03690, 2026.
- [2] G. Lopez, M. Martin, and R. Paya, *Real Banach spaces with numerical index 1*, Bulletin of the London Mathematical Society 31 (1999), 207–212.